

Hyperuniformity in Chemotactic Active Matter

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1 Models

1.1 Particle-based model

$$\frac{d}{dt}\mathbf{r}_i(t) = \alpha \nabla c(\mathbf{r}_i(t), t) + \sqrt{2D}\boldsymbol{\xi}(t) , \quad (1a)$$

$$\frac{\partial}{\partial t}c(\mathbf{r}, t) = D_c \nabla^2 c(\mathbf{r}, t) + \mu c(\mathbf{r}, t) + k \sum_{j=1}^N \delta(\mathbf{r} - \mathbf{r}_j(t)) , \quad (1b)$$

for $i = 1, 2, \dots, N$. Here, $\mathbf{r}_i$ is the position of the i -th particle, c is the concentration field of the chemoattractant, α is the chemotactic sensitivity. If $\alpha > 0$, the particles are attracted to higher concentrations of the chemoattractant (chemoattraction/positive chemotaxis), while if $\alpha < 0$, the particles are repelled by higher concentrations of the chemoattractant (chemorepulsion/negative chemotaxis). D is the diffusion coefficient of the particles, $\boldsymbol{\xi}$ is a Gaussian white noise with zero mean and unit variance, D_c is the diffusion coefficient of the chemoattractant, μ is the decay ($\mu < 0$) or growth ($\mu > 0$) rate of the chemoattractant, and k is the production ($k > 0$) or consumption rate ($k < 0$) of the chemoattractant by the particles.

For simplicity, we assume that particles are point-like and move in a two-dimensional $L \times L$ square domain with periodic boundary conditions. The initial positions of the particles can be randomly distributed, and the initial concentration field can be set to a small random perturbation around a uniform background concentration c_0 .

To reduce the parameter space, we can non-dimensionalize the equations by introducing $x_0 = L$, $t_0 = 1/k$ and c_0 as characteristic scales of length, time and concentration. The resulting dimensionless parameters are $\tilde{\alpha} = \frac{\alpha c_0}{L^2 k}$, $\tilde{D} = \frac{D}{L^2 k}$, $\tilde{D}_c = \frac{D_c}{L^2 k}$, and $\tilde{\mu} = \frac{\mu}{k}$. We can omit the tildes and work with the dimensionless equations:

$$\frac{d}{dt}\mathbf{r}_i(t) = \alpha \nabla c(\mathbf{r}_i(t), t) + \sqrt{2D}\boldsymbol{\xi}(t) , \quad (2a)$$

$$\frac{\partial}{\partial t}c(\mathbf{r}, t) = D_c \nabla^2 c(\mathbf{r}, t) + \mu c(\mathbf{r}, t) + \sum_{j=1}^N \delta(\mathbf{r} - \mathbf{r}_j(t)) . \quad (2b)$$

Note that the above equations mathematically implies the assumption of $k > 0$. If $k < 0$, the non-dimensionalization would be different, and the Eq. (2b) would have a negative sign in front of the summation term:

$$\frac{\partial}{\partial t} c(\mathbf{r}, t) = D_c \nabla^2 c(\mathbf{r}, t) + \mu c(\mathbf{r}, t) - \sum_{j=1}^N \delta(\mathbf{r} - \mathbf{r}_j(t)) . \quad (3)$$

In the following, we will focus on the case of $k < 0$ (chemorepulsion/negative chemotaxis), which is more relevant to the emergence of hyperuniformity.

As shown in the snapshots, the case of production ($k > 0$) leads to aggregation of particles and formation of dense clusters, while the case of consumption ($k < 0$) leads to dispersion of particles and formation of more uniform distributions. The latter case is more likely to exhibit hyperuniformity.

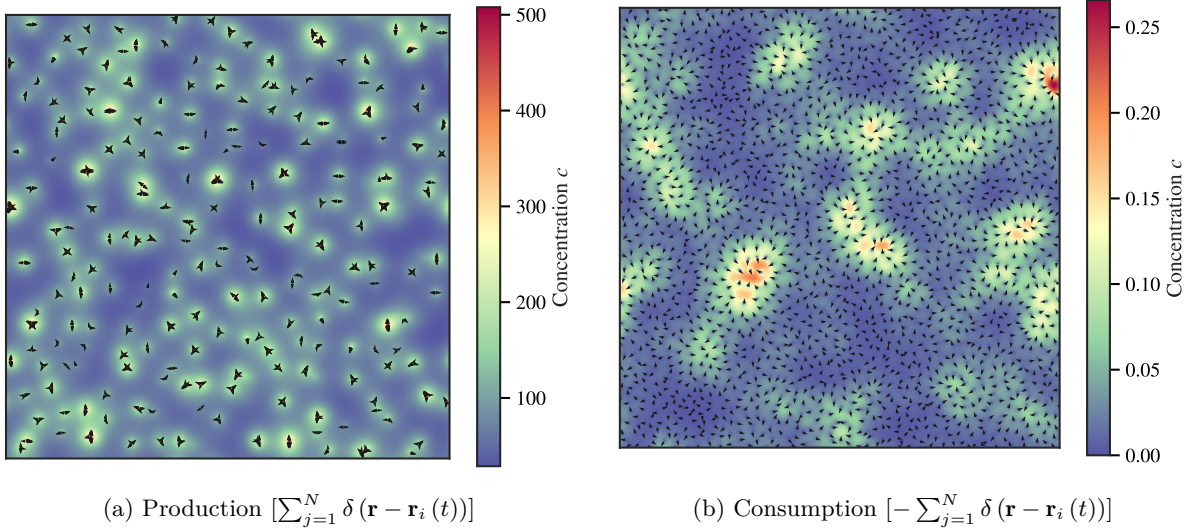


Figure 1: Snapshots of particle configurations for (a) production and (b) consumption of chemoattractant. The parameters are $\alpha = 0.1$, $D = 0.001$, $D_c = 0.1$, $\mu = 0.1$, and $N = 2000$ particles in an $L = 20$ domain. The background color represents the concentration field $c(\mathbf{r}, t)$, with colorbar indicating the concentration level. The positions and velocity direction of the particles are shown as arrows.

1.2 Continuum model

It is convenient to describe collective behavior using a particle density field $\rho(\mathbf{r}, t) = \sum_{i=1}^N \delta(\mathbf{r} - \mathbf{r}_i(t))$ (Note that the $\rho(\mathbf{r}, t)$ here is particle number density, not probability density). By applying the Dean's method [1], we can derive the following stochastic partial differential equations for ρ and c :

$$\frac{\partial}{\partial t} \rho(\mathbf{r}, t) = -\alpha \nabla \cdot (\rho \nabla c) + D \nabla^2 \rho - \nabla \cdot [\sqrt{2D\rho} \mathbf{\Lambda}] , \quad (4a)$$

$$\frac{\partial}{\partial t} c(\mathbf{r}, t) = D_c \nabla^2 c + \mu c - \rho , \quad (4b)$$

where $\mathbf{\Lambda}(\mathbf{r}, t)$ is a Gaussian white noise with zero mean and unit variance, satisfying

$$\langle \Lambda_a(\mathbf{r}, t) \Lambda_b(\mathbf{r}', t') \rangle = \delta_{ab} \delta(\mathbf{r} - \mathbf{r}') \delta(t - t') . \quad (5)$$

Under the mean-field approximation, $\langle \mathbf{\Lambda} \rangle = 0$ leads to $\langle -\nabla \cdot [\sqrt{2D\rho} \mathbf{\Lambda}] \rangle \approx 0$, and we can neglect the noise term. This gives us the classical Keller-Segel model [2, 3]:

$$\frac{\partial}{\partial t} \rho(\mathbf{r}, t) = -\alpha \nabla \cdot (\rho \nabla c) + D \nabla^2 \rho , \quad (6a)$$

$$\frac{\partial}{\partial t} c(\mathbf{r}, t) = D_c \nabla^2 c + \mu c - \rho . \quad (6b)$$

One can perform linear stability analysis on the Keller-Segel model to determine the conditions for pattern formation. However, the mean-field approximation neglects fluctuations, which can be crucial for understanding the emergence of hyperuniformity. Therefore, we need to consider the full stochastic equations with noise to capture the effects of fluctuations on the system's behavior.

One obvious solution of Eqs. (6) is the homogeneous state $\rho(\mathbf{r}, t) = \rho_0$ and $c(\mathbf{r}, t) = c_0$, where $c_0 = \rho_0/\mu$, which can be regarded as solution to Eqs. (4) in the mean-field sense. To analyze the stability of this homogeneous state, we can introduce small perturbations $\delta\rho(\mathbf{r}, t)$ and $\delta c(\mathbf{r}, t)$ around the homogeneous solution:

$$\rho(\mathbf{r}, t) = \rho_0 + \delta\rho(\mathbf{r}, t), \quad c(\mathbf{r}, t) = c_0 + \delta c(\mathbf{r}, t). \quad (7)$$

Introducing these perturbations into Eqs. (4) gives us the following linearized equations:

$$\frac{\partial}{\partial t} \delta\rho = -\alpha\rho_0 \nabla^2 \delta c + D \nabla^2 \delta\rho - \nabla \cdot \left[\sqrt{2D\rho_0} \mathbf{\Lambda} \right], \quad (8a)$$

$$\frac{\partial}{\partial t} \delta c = D_c \nabla^2 \delta c + \mu \delta c - \delta\rho. \quad (8b)$$

By performing Fourier transform on the linearized equations, we can analyze the growth rates of different modes and determine the conditions for instability. The presence of noise can lead to fluctuations that may drive the system towards hyperuniform states, even in regimes where the mean-field analysis predicts stability. Therefore, it is essential to consider the full stochastic dynamics to understand the emergence of hyperuniformity in chemotactic active matter.

We take the Fourier expansion under periodic boundary conditions:

$$\delta\rho(\mathbf{r}, t) = \sum_{\mathbf{k}} \delta\rho_{\mathbf{k}}(t) e^{i\mathbf{k} \cdot \mathbf{r}}, \quad \delta c(\mathbf{r}, t) = \sum_{\mathbf{k}} \delta c_{\mathbf{k}}(t) e^{i\mathbf{k} \cdot \mathbf{r}}. \quad (9)$$

Equivalently, it can also be written as a continuous Fourier transform:

$$\delta\rho_{\mathbf{k}}(t) = \int e^{-i\mathbf{k} \cdot \mathbf{r}} \delta\rho(\mathbf{r}, t) d\mathbf{k}, \quad \delta c_{\mathbf{k}}(t) = \int e^{-i\mathbf{k} \cdot \mathbf{r}} \delta c(\mathbf{r}, t) d\mathbf{k}. \quad (10)$$

The specific normalization factor does not affect the symbol structure here. There is a subtlety in the Fourier transform of the noise term, which can be expressed as:

$$\nabla \cdot \left[\sqrt{2D\rho_0} \mathbf{\Lambda} \right] = \sum_{\mathbf{k}} i\mathbf{k} \cdot \sqrt{2D\rho_0} \mathbf{\Lambda}_{\mathbf{k}}. \quad (11)$$

After Fourier transform, the linearized equations become:

$$\frac{\partial}{\partial t} \delta\rho_{\mathbf{k}} = -Dk^2 \delta\rho_{\mathbf{k}} + \alpha\rho_0 k^2 \delta c_{\mathbf{k}} + \eta_{\mathbf{k}}, \quad (12a)$$

$$\frac{\partial}{\partial t} \delta c_{\mathbf{k}} = -(D_c k^2 - \mu) \delta c_{\mathbf{k}} - \delta\rho_{\mathbf{k}}, \quad (12b)$$

where $\eta_{\mathbf{k}} = -i\mathbf{k} \cdot \sqrt{2D\rho_0} \mathbf{\Lambda}_{\mathbf{k}}$. The noise term $\eta_{\mathbf{k}}$ has zero mean and its correlation can be calculated as follows:

$$\langle \eta_{\mathbf{k}}(t) \eta_{\mathbf{k}'}(t') \rangle = 2D\rho_0 k^2 \delta(t - t'). \quad (13)$$

1.2.1 Fast chemical field approximation

In the limit of fast chemical diffusion, we can assume that the chemical field c quickly reaches a quasi-steady state, which allows us to eliminate c from the equations. Setting $\frac{\partial}{\partial t} \delta c_{\mathbf{k}} \approx 0$ in Eq. (12) gives us:

$$\delta c_{\mathbf{k}} \approx -\frac{\delta\rho_{\mathbf{k}}}{D_c k^2 - \mu}. \quad (14)$$

Substituting this back into the equation for $\delta\rho_{\mathbf{k}}$ yields:

$$\partial_t \delta\rho_{\mathbf{k}} = -k^2 \left[D + \frac{\alpha\rho_0}{D_c k^2 - \mu} \right] \delta\rho_{\mathbf{k}} + \eta_{\mathbf{k}}. \quad (15)$$

Define effective diffusion coefficient $D_{\text{eff}}(k) = D + \frac{\alpha\rho_0}{D_c k^2 - \mu}$, we can rewrite the equation as:

$$\partial_t \delta\rho_{\mathbf{k}} = -k^2 D_{\text{eff}}(k) \delta\rho_{\mathbf{k}} + \eta_{\mathbf{k}} . \quad (16)$$

This equation is an Ornstein-Uhlenbeck process, and we can calculate the steady-state structure factor, which is the variance of density fluctuations at wavevector $\mathbf{k}$, using the formula for the variance of an Ornstein-Uhlenbeck process (see Appendix 2.1):

$$S(k) = \langle \delta\rho_{\mathbf{k}} \delta\rho_{-\mathbf{k}} \rangle = \frac{2D\rho_0 k^2}{2k^2 D_{\text{eff}}(k)} = \frac{D\rho_0}{D + \frac{\alpha\rho_0}{D_c k^2 - \mu}} . \quad (17)$$

Here, the structure factor $S(k)$ can be interpreted as a noise-relaxation balance. The numerator $2D\rho_0 k^2$ represents the strength of the noise driving the fluctuations, while the denominator $2k^2 D_{\text{eff}}(k)$ represents the relaxation rate of the fluctuations due to diffusion and chemotactic interactions.

Let

$$A_0 = D - \frac{\alpha\rho_0}{\mu} . \quad (18)$$

For $k \rightarrow 0$, $\frac{D_c}{|\mu|} k^2 \rightarrow 0$, so we can expand the denominator in a Taylor series:

$$\begin{aligned} \frac{1}{-\mu + D_c k^2} &= -\frac{1}{\mu} \frac{1}{1 - \frac{D_c}{\mu} k^2} \\ &= -\frac{1}{\mu} \left[1 + \frac{D_c}{\mu} k^2 + \left(\frac{D_c}{-\mu} \right)^2 k^4 + \dots \right] , \end{aligned} \quad (19)$$

then

$$D_{\text{eff}}(k) = D - \frac{\alpha\rho_0}{\mu} - \frac{\alpha\rho_0 D_c}{\mu^2} k^2 - \dots , \quad (20)$$

and the structure factor can be approximated as:

$$\begin{aligned} S(k) &= \rho_0 \frac{D}{A_0} \left[1 + \frac{\alpha\rho_0 D_c}{|\mu|^2 A_0} k^2 + \dots \right] \\ &= S_0 + B k^2 + \dots , \end{aligned} \quad (21)$$

where $S_0 = \frac{D\rho_0}{A_0}$ and $B = \frac{\alpha\rho_0^2 D D_c}{|\mu|^2 A_0^2}$.

2 Appendix

2.1 Calculation of variance for Ornstein-Uhlenbeck process

For an Ornstein-Uhlenbeck process described by the equation

$$\partial_t x(t) = -\lambda x(t) + \eta(t) , \quad (22)$$

where $\eta(t)$ is a Gaussian white noise with zero mean and correlation $\langle \eta(t) \eta(t') \rangle = 2\Gamma \delta(t - t')$, the formal solution is

$$X(t) = X(0) e^{-\lambda t} + \int_{-\infty}^t e^{-\lambda(t-s)} \eta(s) ds . \quad (23)$$

After a long time, the initial condition becomes irrelevant:

$$X(t) \approx \int_{-\infty}^t e^{-\lambda(t-s)} \eta(s) ds . \quad (24)$$

The variance of $X(t)$ can be calculated as:

$$\begin{aligned} \langle X^2(t) \rangle &= \int_{-\infty}^t \int_{-\infty}^t e^{-\lambda(2t-s-s')} \langle \eta(s) \eta(s') \rangle ds ds' \\ &= 2\Gamma \int_{-\infty}^t e^{-2\lambda(t-s)} ds \\ &= \frac{\Gamma}{\lambda} . \end{aligned} \quad (25)$$

References

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- [3] Evelyn F. Keller and Lee A. Segel. Initiation of slime mold aggregation viewed as an instability. *Journal of Theoretical Biology*, 26(3):399–415, 1970.